

VPC-Task 1

1.Create VPC with 2 private and 2 public subnets.

Step 1: Create VPC

1. Open AWS Console → VPC.
2. Click Your VPCs → Create VPC.
3. Select VPC only.
4. Name: `DevOps-VPC`
5. IPv4 CIDR: `10.0.0.0/16`
6. Tenancy: Default
7. Click Create VPC.

Step 2: Create Public Subnet 1

1. Go to Subnets → Create subnet.
2. VPC: `DevOps-VPC`
3. Name: `Public-Subnet-1`
4. AZ: `eu-north-1a`
5. CIDR: `10.0.1.0/24`
6. Click Create subnet.

Step 3: Create Public Subnet 2

1. Create subnet
2. VPC: `DevOps-VPC`
3. Name: `Public-Subnet-2`
4. AZ: `eu-north-1b`
5. CIDR: `10.0.2.0/24`
6. Create subnet

Step 4: Create Private Subnet 1

1. Create subnet
2. VPC: `DevOps-VPC`
3. Name: `Private-Subnet-1`
4. AZ: `eu-north-1a`
5. CIDR: `10.0.3.0/24`
6. Create subnet

Step 5: Create Private Subnet 2

1. Create subnet
2. VPC: DevOps-VPC
3. Name: Private-Subnet-2
4. AZ: eu-north-1b
5. CIDR: 10.0.4.0/24
6. Create subnet

Subnets (4) Info

Find subnets by attribute or tag

Subnet ID: subnet-07631a1143296ce73 X Subnet ID: subnet-0231e37fbefc0cf81 X Subnet ID: subnet-0eb8ba292c9ba4766 X (+1)

Clear filters

☐	Name	Subnet ID	State	VPC	Block Public...	IP
☐	Public-Subnet-2	subnet-0231e37fbefc0cf81	Available	vpc-0ee80a67f9de88fa7 DevO...	Off	10
☐	Private-Subnet-2	subnet-0e292c9b518a64943	Available	vpc-0ee80a67f9de88fa7 DevO...	Off	10
☐	Public-Subnet-1	subnet-07631a1143296ce73	Available	vpc-0ee80a67f9de88fa7 DevO...	Off	10
☐	Private-Subnet-1	subnet-0eb8ba292c9ba4766	Available	vpc-0ee80a67f9de88fa7 DevO...	Off	10

2. Enable DNS Hostname in VPC.

1. Open **AWS Console** → **VPC**.
2. Click **Your VPCs**.
3. Select **DevOps-VPC**.
4. Click **Actions** → **Edit VPC settings**.
5. Find **DNS settings**.
6. Enable/check **DNS hostnames** ☒
7. Click **Save changes**.

Edit VPC settings Info

VPC details

VPC ID: vpc-0ee80a67f9de88fa7

Name: DevOps-VPC

DHCP settings

DHCP option set: dopt-066d2f9d97b51d401

DNS settings

☒ Enable DNS resolution Info

☒ Enable DNS hostnames Info

Network Address Usage metrics settings

☐ Enable Network Address Usage metrics Info

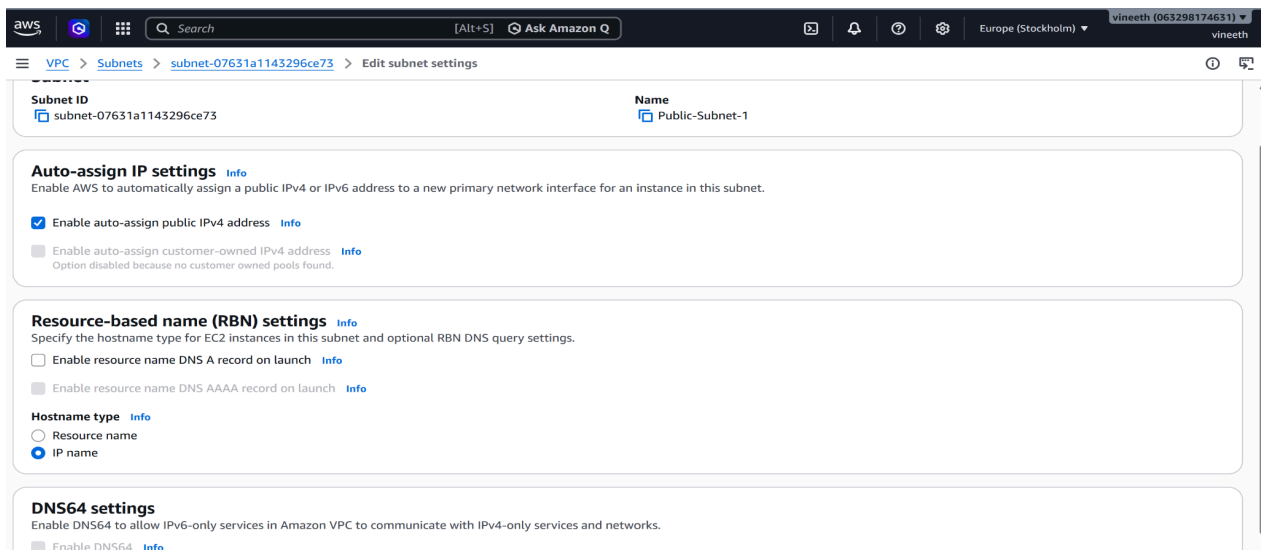
Cancel Save

3. Enable Auto Assign Public IP in 2 public subnets.

Public-Subnet-1

1. Go to **VPC** → **Subnets**.
2. Select **Public-Subnet-1**.
3. **Actions** → **Edit subnet settings**.
4. Enable ☒ **Auto-assign public IPv4 address**.
5. Click **Save**.

☒ **Check:** Both public subnets should show **Auto-assign Public IPv4 = Yes/Enabled**.



Public-Subnet-2

1. Select **Public-Subnet-2**.
2. **Actions** → **Edit subnet settings**.
3. Enable ☒ **Auto-assign public IPv4 address**.
4. Click **Save**.

aws [Search] [Alt+S] Ask Amazon Q Europe (Stockholm) vineeth (063298174631) vineeth

VPC > Subnets > subnet-0231e37fbefc0cf81 > Edit subnet settings

Edit subnet settings [Info](#)

Subnet

Subnet ID [🔗](#) subnet-0231e37fbefc0cf81

Name [🔗](#) Public-Subnet-2

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch [Info](#)

☐ Enable resource name DNS AAAA record on launch [Info](#)

Hostname type [Info](#)

☐ Resource name

☒ IP name

4. Add 2 private subnets in private route table

1. Go to **VPC** → **Route Tables**.
2. Click **Create route table**.
3. Name: **Private-Route-Table**
4. Select VPC: **DevOps-VPC**
5. Click **Create route table**.
6. Select **Private-Route-Table**.
7. Go to **Subnet associations** → **Edit subnet associations**.
8. Select ☒ **Private-Subnet-1** and ☒ **Private-Subnet-2**.
9. Click **Save associations**.

VPC > Route tables > rtb-01b46d70bade565f1

VPC dashboard [Filter by VPC:](#) [AWS Global View](#)

Virtual private cloud

- Your VPCs
- Subnets
- [Route tables](#)
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets
- Elastic IPs

Details [Info](#)

Route table ID [🔗](#) rtb-01b46d70bade565f1

VPC [🔗](#) vpc-0ee80a67f9de88fa7 | DevOps-VPC

Main [🔗](#) No

Owner ID [🔗](#) 063298174631

Explicit subnet associations [2 subnets](#)

Edge associations [-](#)

[Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#) [Related resources](#)

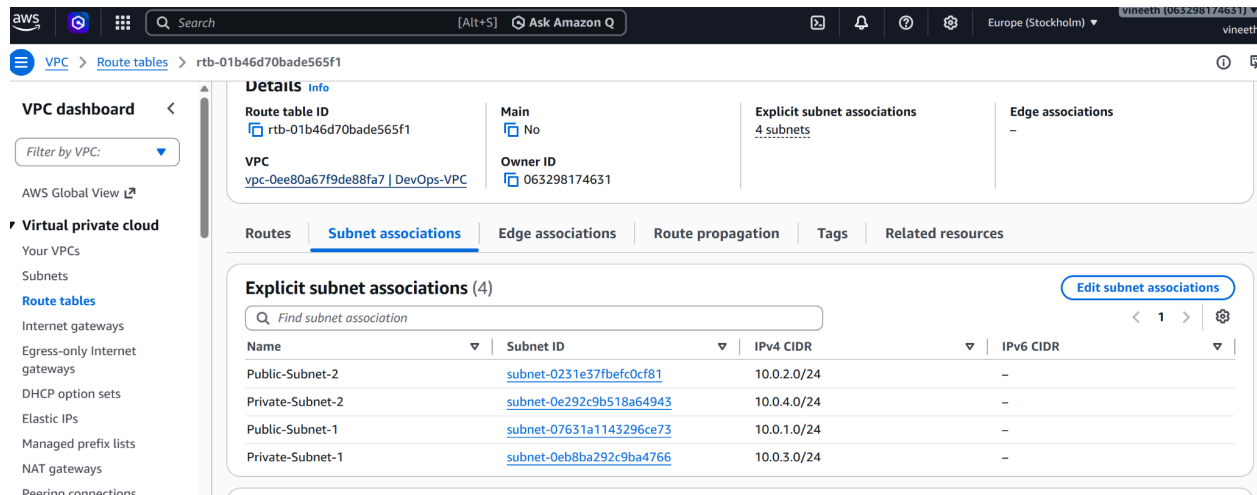
Explicit subnet associations (2) [Edit subnet associations](#)

[Find subnet association](#)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Private-Subnet-2	subnet-0e292c9b518a64943	10.0.4.0/24	-
Private-Subnet-1	subnet-0eb8ba292c9ba4766	10.0.3.0/24	-

5. Add 2 public subnets in public route table

1. Go to **AWS Console** → **VPC** → **Route Tables**.
2. Click **Create route table**.
3. Name: **Public-Route-Table**
4. Select VPC: **DevOps-VPC**
5. Click **Create route table**.
6. Select **Public-Route-Table**.
7. Go to **Subnet associations** → **Edit subnet associations**.
8. Select ☒ **Public-Subnet-1**
9. Select ☒ **Public-Subnet-2**
10. Click **Save associations**.



6. Public route table will have the routes to internet and local.

Add Internet and Local Routes to Public Route Table

1. Go to **AWS Console** → **VPC** → **Route Tables**.
2. Select **Public-Route-Table**.
3. Go to **Routes** → **Edit routes**.
4. You should already have:
 - o **10.0.0.0/16** → **local** ☒
5. Click **Add route**.
6. Destination: **0.0.0.0/0**

7. Target: **Internet Gateway (IGW)** → select your IGW.
8. Click **Save changes**.

VPC dashboard <

Filter by VPC: ▾

AWS Global View ⓘ

▼ Virtual private cloud

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

▼ Security

Updated routes for rtb-08280530dd96cae68 / Public-Route-Table successfully

Details

rtb-08280530dd96cae68 / Public-Route-Table Actions ▾

Details Info

Route table ID rtb-08280530dd96cae68	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-0ee80a67f9de88fa7	Owner ID 063298174631		

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2) Both ▾ Edit routes

Filter routes

Destination ▾	Target ▾	Status ▾	Propagated ▾	Route Origin ▾
0.0.0.0/0	igw-039b37297ed2802e3	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

7. Create EC2 in public subnet with t2.micro and install PHP

Create EC2 and Install PHP

1. **EC2** → **Instances** → **Launch Instance**
2. Name: **PHP**
3. AMI: **Amazon Linux 2023**
4. Instance type: **t2.micro**
5. Select **Key Pair**
6. Network → **VPC: DevOps-VPC**
7. **Subnet: Public-Subnet-1**
8. **Auto-assign Public IP: Enable**
9. Security Group:
 - SSH → 22 → My IP
 - HTTP → 80 → Anywhere
10. Click **Launch Instance**
11. Select EC2 → **Connect** → **EC2 Instance Connect** → **Connect**
12. Install PHP:

Warning: Permanently added '10.0.3.46' (ED25519) to the list of known hosts.



Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

```
[ec2-user@ip-10-0-3-46 ~]$ ping google.com
PING google.com (172.217.19.238) 56(84) bytes of data.
64 bytes from par21s11-in-f14.1e100.net (172.217.19.238): icmp_seq=1 ttl
=117 time=4.08 ms
64 bytes from par21s11-in-f14.1e100.net (172.217.19.238): icmp_seq=2 ttl
=117 time=3.79 ms
64 bytes from par21s11-in-f14.1e100.net (172.217.19.238): icmp_seq=3 ttl
=117 time=3.79 ms
64 bytes from par21s11-in-f14.1e100.net (172.217.19.238): icmp_seq=4 ttl
=117 time=4.33 ms
64 bytes from par21s11-in-f14.1e100.net (172.217.19.238): icmp_seq=5 ttl
=117 time=3.80 ms
^Z
[1]+  Stopped                  ping google.com
```

9: Install Apache Tomcat in private EC2 and deploy a sample app.

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	private-ec2	i-0328f47dfbf246de	Running	t3.micro	3/3 checks passed	eu-north-1a	ec2-51-20-181-190
☐	Public-ec2	i-026f0f412c232de13	Running	t3.micro	3/3 checks passed	eu-north-1a	ec2-13-61-141-228

```

apache-tomcat-10.1.59/webapps/host-manager/WEB-INF/jsp/401.jsp
apache-tomcat-10.1.59/webapps/host-manager/WEB-INF/jsp/403.jsp
apache-tomcat-10.1.59/webapps/host-manager/WEB-INF/jsp/404.jsp
apache-tomcat-10.1.59/webapps/host-manager/WEB-INF/manager.xml
apache-tomcat-10.1.59/webapps/host-manager/WEB-INF/web.xml
apache-tomcat-10.1.59/webapps/host-manager/css/manager.css
apache-tomcat-10.1.59/webapps/host-manager/images/asf-logo.svg
apache-tomcat-10.1.59/webapps/host-manager/images/favicon.ico
apache-tomcat-10.1.59/webapps/host-manager/images/tomcat.svg
apache-tomcat-10.1.59/webapps/host-manager/index.jsp
apache-tomcat-10.1.59/webapps/manager/META-INF/context.xml
apache-tomcat-10.1.59/webapps/manager/WEB-INF/jsp/401.jsp
apache-tomcat-10.1.59/webapps/manager/WEB-INF/jsp/403.jsp
apache-tomcat-10.1.59/webapps/manager/WEB-INF/jsp/404.jsp
apache-tomcat-10.1.59/webapps/manager/WEB-INF/jsp/connectorCerts.jsp
apache-tomcat-10.1.59/webapps/manager/WEB-INF/jsp/connectorCiphers.jsp
apache-tomcat-10.1.59/webapps/manager/WEB-INF/jsp/connectorTrustedCerts.jsp
apache-tomcat-10.1.59/webapps/manager/WEB-INF/jsp/sessionDetail.jsp
apache-tomcat-10.1.59/webapps/manager/WEB-INF/jsp/sessionsList.jsp
apache-tomcat-10.1.59/webapps/manager/WEB-INF/web.xml
apache-tomcat-10.1.59/webapps/manager/css/manager.css
apache-tomcat-10.1.59/webapps/manager/images/asf-logo.svg
apache-tomcat-10.1.59/webapps/manager/images/favicon.ico
apache-tomcat-10.1.59/webapps/manager/images/tomcat.svg
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ sudo vi /opt/apache-tomcat-10.1.59/webapps/sampleapp/index.html
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ curl http://localhost:8080/sampleapp/
[doctype html>html lang=en> <head><title>HTTP Status 404 - Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 - Not Found</h1><hr class="line" /><p><b>Type</b>: Status Report</p><p><b>Message</b>: The requested resource [/sampleapp/47/] is not available</p><p><b>Description</b>: The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /></body></html>[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ sudo vi /opt/apache-tomcat-10.1.59/webapps/sampleapp/index.html
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ curl http://localhost:8080/sampleapp/
[doctype html>html lang=en> <head><title>HTTP Status 404 - Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 - Not Found</h1><hr class="line" /><p><b>Type</b>: Status Report</p><p><b>Message</b>: The requested resource [/sampleapp/47/] is not available</p><p><b>Description</b>: The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /></body></html>[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ cd ..
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ echo "chiHello from sampleapp\<h1>" | sudo tee /opt/apache-tomcat-10.1.59/webapps/sampleapp/index.html
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ curl http://localhost:8080/sampleapp/
[doctype html>html lang=en> <head><title>HTTP Status 404 - Not Found</title><style type="text/css">body {font-family:Tahoma,Arial,sans-serif;} h1, h2, h3, b {color:white;background-color:#525D76;} h1 {font-size:22px;} h2 {font-size:16px;} h3 {font-size:14px;} p {font-size:12px;} a {color:black;} .line {height:1px;background-color:#525D76;border:none;}</style></head><body><h1>HTTP Status 404 - Not Found</h1><hr class="line" /><p><b>Type</b>: Status Report</p><p><b>Message</b>: The requested resource [/sampleapp/47/] is not available</p><p><b>Description</b>: The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.</p><hr class="line" /></body></html>[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ cd ..
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ cd /opt
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ cd ..
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ sudo vi /opt/apache-tomcat-10.1.59/webapps/sampleapp/index.html
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ sudo vi webapps/sampleapp/index.html
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ cd ..
[ec2-user@ip-10-0-3-46 apache-tomcat-10.1.59]$ sudo vi /opt/apache-tomcat/webapps/sampleapp/index.html

```

10: Configure VPC flow logs and store the logs in S3 and CloudWatch.

Part 1 — Flow Logs → S3

1. VPC → Your VPCs → DevOps-VPC
2. Flow logs → Create flow log
3. Name: DevOps-FlowLog-S3
4. Filter: All
5. Aggregation: 10 minutes
6. Destination: S3
7. Enter S3 ARN: `arn:aws:s3:::task-vpc-flow-logs-929523583693`
8. Click Create flow log 

Part 2 — Flow Logs → CloudWatch

1. VPC → Your VPCs → DevOps-VPC
2. Flow logs → Create flow log
3. Name: DevOps-FlowLog-CloudWatch
4. Filter: All
5. Aggregation: 10 minutes
6. Destination: CloudWatch Logs
7. Log group: `/aws/vpc/DevOps-flow-logs`

8. **Create and use a new service role**
9. Role: **VPCFlowLogs-CloudWatch**
10. Click **Create flow log** ✓

Final: Both flow logs should show **Active**.

The screenshot shows the AWS VPC console interface. A green banner at the top indicates: "Successfully created flow log for the following resource: vpc-0ee80a67f9de88fa7". The main heading is "fl-07805aed084ba9b8c / DevOps-VPC-FlowLog-CloudWatch".

Details:

Flow log ID fl-07805aed084ba9b8c	Destination Type s3	Traffic Type All	File Format Plain text
Name DevOps-VPC-FlowLog-CloudWatch	Destination Name task-vpc-flow-logs-929523583693	Max Aggregation Interval 10 minutes	Hive Compatible Partitions Not enabled
State Active	IAM Role -	Log Format Default	Partition Logs Daily
Creation Time Tuesday, September 8, 2026 at 23:48:39 GMT+5:30	Cross Account IAM Role -	Tag field specifications -	

Below the details are tabs for "Tags" and "Integrations".

The screenshot shows the AWS VPC console interface. A green banner at the top indicates: "Successfully created flow log for the following resource: vpc-0ee80a67f9de88fa7". The main heading is "fl-0e1d975dd4e3eecfa / DevOps-FlowLog-CloudWatch".

Details:

Flow log ID fl-0e1d975dd4e3eecfa	Destination Type cloud-watch-logs	Traffic Type All	File Format -
Name DevOps-FlowLog-CloudWatch	Destination Name /aws/vpc/DevOps-flow-logs	Max Aggregation Interval 10 minutes	Hive Compatible Partitions -
State Active	IAM Role arn:aws:iam::063298174631:role/service-role/VPCFlowLogs-CloudWatch	Log Format Default	Partition Logs -
Creation Time Tuesday, September 8, 2026 at 23:53:50 GMT+5:30	Cross Account IAM Role -	Tag field specifications -	

Below the details are tabs for "Tags" and "Integrations".

Tags:

Search tags:

Manage tags: [Manage tags](#)

